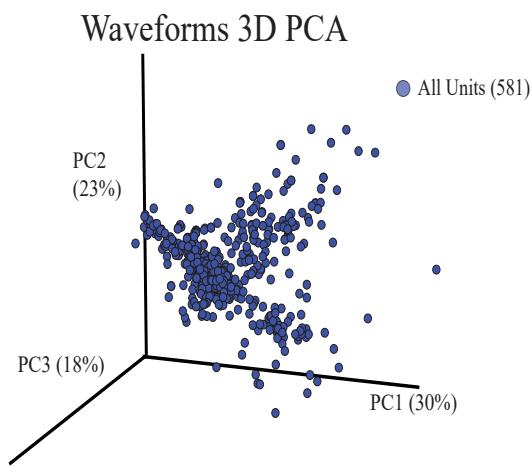


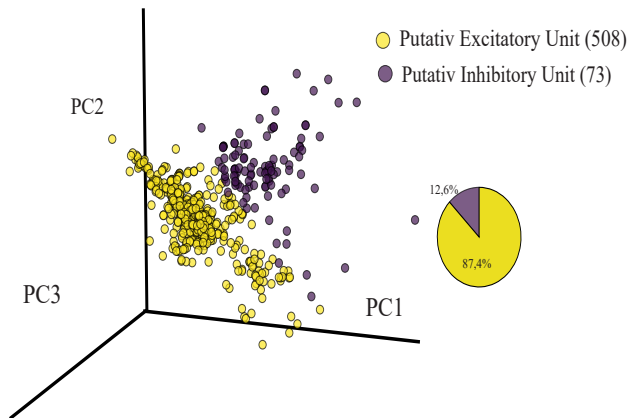
Figure 5

A

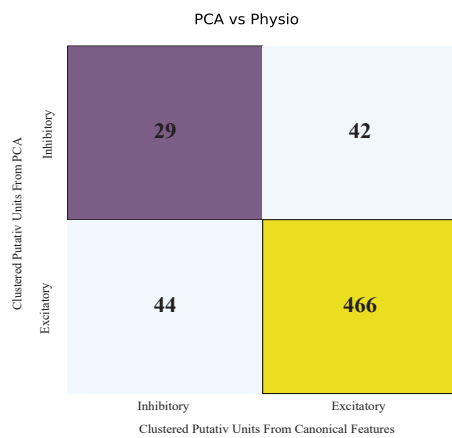


B

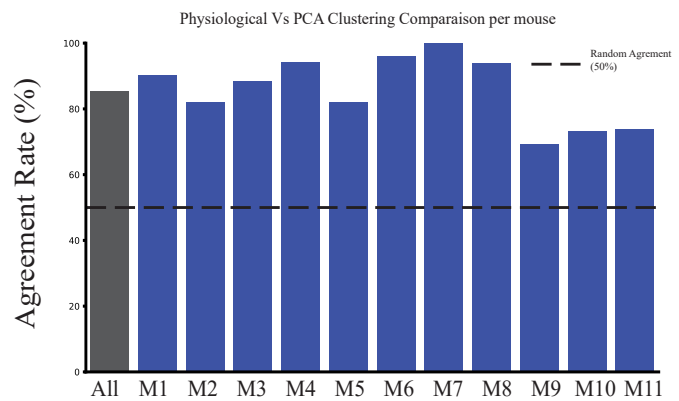
K-Means on PCA of Full Waveform Features



C

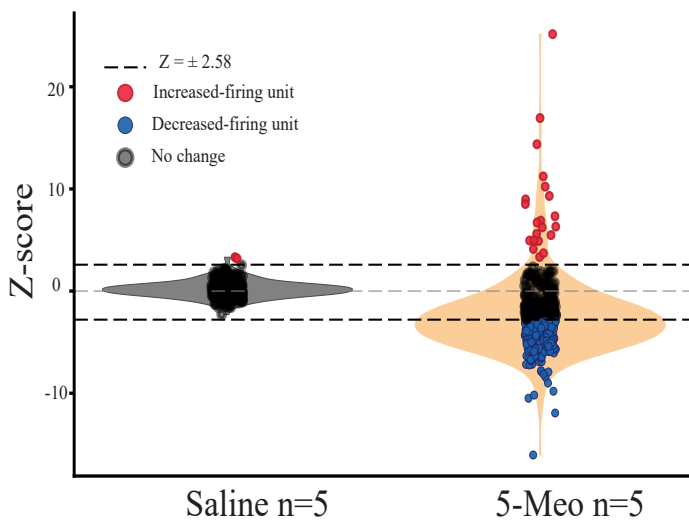


D



E

Unit Response Distribution



F

Proportion of Modulated Units

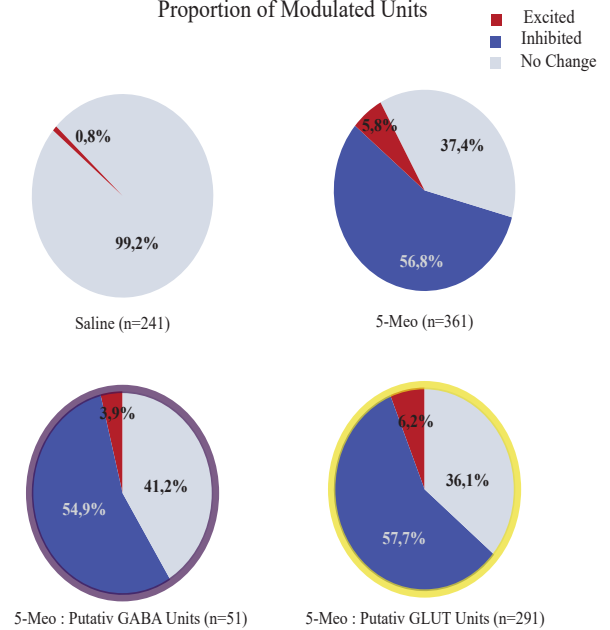


Figure 5. Single-unit classification and modulation of neuronal firing dynamics under 5-MeO-DMT.

A. Principal component analysis (PCA) of waveform features across all recorded units. Scatterplot of units ($n = 581$) in the space of the first three principal components derived from the waveform feature matrix. Each point represents one unit, positioned according to its scores on PC1 (30% variance explained), PC2 (23%), and PC3 (18%). PCA was applied to reduce dimensionality and visualize structure in the electrophysiological diversity of the recorded population. **C–D. K-means clustering of single units based on waveform features and PCA projections.**

B. Clustering Scatterplot of units ($n = 581$) in a 3D space defined by canonical waveform-derived features: half-width duration, peak-to-trough ratio, and mean firing rate. Units were clustered using k -means ($k = 2$), yielding two populations labeled as putative excitatory (black) and inhibitory (yellow) based on waveform geometry.

C. Confusion matrix comparing unit labels obtained via PCA-based clustering of waveform features versus physiologically defined categories (based on canonical waveform metrics). Each cell indicates the number of units classified into each combination of categories across all recorded units ($n = 581$). The overall agreement between methods is 85.2%, with 466 units classified as excitatory and 29 as inhibitory by both methods.

D. Agreement rate per animal between PCA- and physiology-based classifications. Each point represents one mouse; the vertical axis shows the percentage of units with matching classification labels across methods. Bars indicate mean $\pm$ standard deviation across mice ($85.6\% \pm 9.0\%$). Agreement values range from 69.2% to 100%, indicating consistency at both the group and individual levels.

E. Distribution of mean post-injection firing rate changes across single units. Violin plots show the distribution of mean Z-scored firing rates (relative to baseline) for all units recorded in saline ($n = 5$ mice) and 5-MeO-DMT ($n = 5$ mice) groups. Each point represents one unit. Units with a post-injection Z-score $> +2.58$ (red) are classified as increased-firing; those with $Z < -2.58$ (blue) are classified as decreased-firing; units within this range (gray) show no significant change. Dashed lines indicate $Z = \pm 2.58$ threshold corresponding to a two-tailed $p < 0.01$. Violin width reflects kernel density

F. Proportions of modulated units following 5-MeO-DMT administration. Pie charts show the distribution of unit modulation types based on post-injection firing rate Z-scores: excited (red, $Z > +2.58$), inhibited (blue, $Z < -2.58$), and non-modulated (gray, $-2.58 < Z < +2.58$). Top panels: comparison of all units recorded in the saline group ($n = 241$ units, 5 mice) and 5-MeO-DMT group ($n = 361$ units, 5 mice). Bottom panels: modulation proportions in the 5-MeO-DMT group, separated by unit type based on waveform classification—putative GABAergic ($n = 51$) and putative glutamatergic units ($n = 291$). Significance of intergroup differences was tested using chi-squared or Mann–Whitney tests, as indicated. Asterisks denote levels of statistical significance ($p < 0.05$, $p < 0.01$, $p < 0.001$; n.s. = not significant). Percent values are indicated within each pie sector.